

CMT1300 High Reliability Enhanced Isolation Amplifier

Characteristic

- Isolation voltage up to 5000Vrms
- ± 50 mV linear input voltage range
- Fixed gain: 41
- Low offset error and drift:
CMT1300D05: ± 0.1 mV (Maximum value),
 $-0.8 \sim 1$ μ V/°C (Maximum)
- Low gain error and drift:
 $\pm 0.3\%$ (max), ± 50 ppm/°C(max)
- Low nonlinearity and drift:
 $\pm 0.03\%$ (max), ± 1 ppm/°C(typical)
- Signal to noise ratio: 86 dB (typical, BW =10 kHz),
72 dB (typical, BW =100 kHz)
- Wide bandwidth: 310 kHz (typical)
- High CMTI: 150 kV/ μ s (typical)
- System-level diagnostic functions:
VDD1 Monitoring
Input common mode overvoltage detection
- Operating temperature: $-40^{\circ}\text{C} \sim 125^{\circ}\text{C}$
- Complies with AEC-Q100 Automotive Application Standard
- UL Approved: 5000 Vrms isolation for 1 minute per UL1577 standard
- CQC certification in accordance with GB4943.1-2022
- CSA Component 5A Certification IEC60950-1 Standard
- DIN VDE V 0884-11:20 17-01
- RoHS Compliant Package: SOP8 (300 mil)

Application

- Shunt current monitoring
- AC motor control
- Uninterruptible power supply
- Car charger

Description

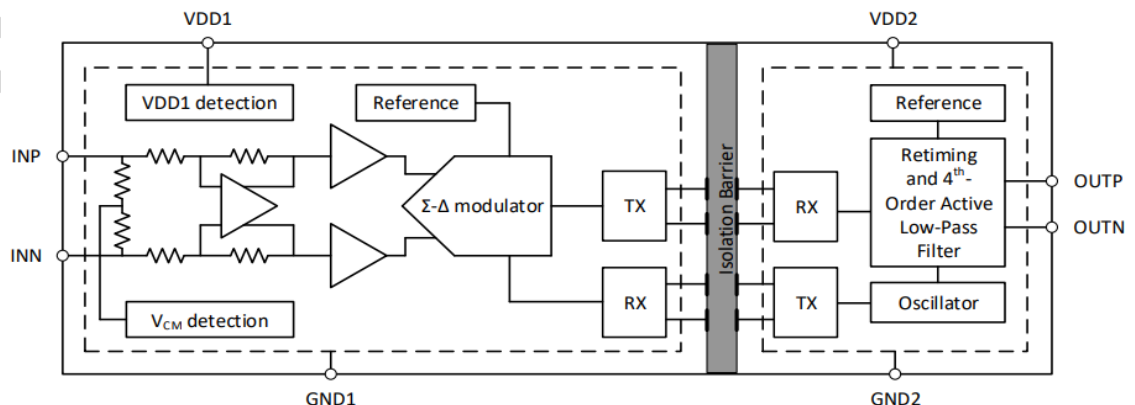
CMT1300 is a high-performance enhanced isolation amplifier based on HOPERF's capacitive isolation technology to isolate the output from the input. The linear differential input signal range of this device is ± 50 mV (± 64 mV full scale). Differential inputs are well-suited for shunt resistor-based current sensing in high-voltage applications that require electrical isolation. The device has a fixed gain (41) and provides a differential analog output.

Low offset and gain drift ensure accuracy over the temperature range. High common mode transient immunity (CMTI) ensures that the device can provide accurate and reliable detection even in the presence of high-power switches in motor control applications.

Fail-safe functions include input common-mode overvoltage detection and VDD1 loss detection, simplifying system-level design and diagnostics.

Order Information

Product Model	Package	Minimum Order Quantity
CMT1300D05	SOW8 (300 mil)	1000



Functional Block Diagram

Catalogue

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1 Pin Description

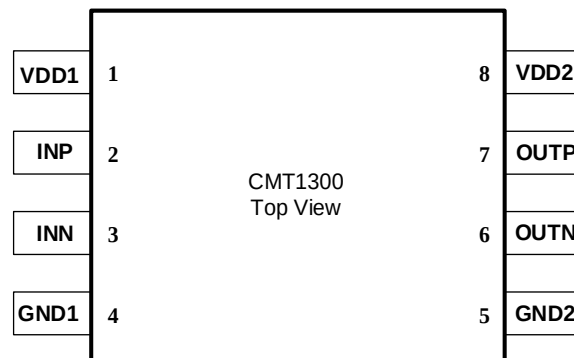


Table 1-1. CMT1300 Pin Description

Pin No.	Symbol	Function
1	VDD1	Power supply on isolator side 1 (3.0 V to 5.5 V)
2	INP	In-phase analog input (CMT1300D05 recommended ± 50 mV)
3	INN	Inverting analog input
4	GND1	GND1, ground reference for isolator side 1
5	GND2	GND2, ground reference for isolator side 2
6	OUTN	Inverted analog output
7	OUTP	In-phase analog output
8	VDD2	Power supply on isolator side 2 (3.0 V to 5.5 V)

2 Specifications

2.1 Recommended Work Rating

Table 2-1. Recommended Operating Ratings

Parameter		Symbol	Min	Typ	Max.	Unit
Side1 supply voltage		VDD1	3.0	5.0	5.5	V
Side 2 supply voltage		VDD2	3.0	3.3	5.5	V
CMT1300D05	Differential input voltage before output clipping	$V_{Clipping}$		± 64		mV
	Linear differential input full scale voltage	V_{FSR}	-50		50	mV
	Input common mode voltage range	V_{CM}	-0.032		0.8	V
Operating ambient temperature		T_A	-40		125	$^{\circ}\text{C}$

2.2 Absolute Maximum Rating

Table 2-1. Absolute Maximum Rating^[1]

Parameter	Symbol	Min	Typ	Max	Unit
Power supply voltage	VDD1, VDD2	-0.3		6.5	V
Input voltage	INP, INN	GND1-6		VDD1+0.5	V
Output voltage	Output, output	GND2-0.5		VDD2+0.5	V
Output current of each output pin	I _O	-10		10	mA
Operating temperature	T _{OPR}	-40		125	°C
Junction temperature	T _J	-40		150	°C
Storage temperature	T _{STG}	-55		150	°C
Electrostatic discharge	HBM ⁽¹⁾	±2000			V
	CDM ⁽²⁾	±1000			V
(1) Human Body Model (HBM), according to AEC-Q100-002-Rev D (2) Charged Device Model (CDM), according to AEC-Q100-011-Rev B					

2.3 Thermal Information

Table 2-3. Thermal Information

Parameter	Symbol	CMT1300	Unit
		SOW8	
Thermal resistance from junction to ambient	R _{θJA}	86	°C/W
Thermal resistance from junction to case (top)	R _{θJC(top)}	28	°C/W
Thermal resistance from junction to board	R _{θJB}	42	°C/W
Junction-to-top characterization parameters	ψ _{JT}	4	°C/W
Junction-to-board characterization parameters	ψ _{JB}	42	°C/W

2.4 Insulation parameters

2.4.1 Insulation and Safety-related Specifications

Parameter	Symbol	Value	Unit	Description
Minimum external air gap (clearance)	CLR	8	mm	Shortest port-to-port air distance
Minimum external path (creepage distance)	CPG	8	mm	Shortest end-to-end Distance across the package surface
Minimum internal clearance	DTI	20	μm	Distance through insulation

Relative leakage index	CTI	>600	V	DIN EN 60112 (VDE 0303-11); IEC 60112
Material group		I		IEC 60664-1

2.4.2 Insulation Characteristics

Description	Test Condition	Symbol	Value	Unit
DIN VDE 0110				
Installation classification according to DIN VDE 0110/1.89	For the rated main voltage ≤ 150 V_{rms}		I to IV	
	For the rated main voltage ≤ 300 V_{rms}		I to IV	
	For the rated main voltage ≤ 400 V_{rms}		I to IV	
Climate classification			40/125/21	
Degree of pollution in accordance with DIN VDE 0110, Table 1			2	
Maximum repeated isolation voltage		V_{IORM}	2121	V_{PEAK}
Maximum operating isolation voltage	AC voltage	V_{IOWM}	1500	V_{RMS}
	DC voltage		2121	V_{DC}
Input to output test voltage, Method B1	$V_{IORM} \times 1.875 = V_{pd(m)}$, 100% production test $t_{ini} = t_m = 1$ sec, Partial discharge < 5 pC	$V_{pd(m)}$	3977	V_{PEAK}
Input to output test voltage, Method A				
Group 1 after environmental testing	$V_{IORM} \times 1.6 = V_{pd(m)}$, $t_{ini} = 60$ sec, $t_m = 10$ sec, Partial discharge < 5 pC	$V_{pd(m)}$	3394	V_{PEAK}
Group 2 and Group 3 after input and safety testing	$V_{IORM} \times 1.2 = V_{pd(m)}$, $t_{ini} = 60$ sec, $t_m = 10$ sec, Partial discharge < 5 pC	$V_{pd(m)}$	2545	V_{PEAK}
Maximum transient isolation voltage	$t = 60$ sec	V_{IOTM}	8000	V_{PEAK}
Maximum surge isolation voltage	Test method according to IEC60065, 1.2/50us waveform, $V_{TEST} = V_{IOSM} \times 1.6$	V_{IOSM}	6250	V_{PEAK}
Insulation resistance	$V_{IO} = 500V$, $T_{amb} = T_s$	R_{IO}	$> 10^9$	Ω
	$V_{IO} = 500V$, $100^\circ C \leq T_{amb} \leq 125^\circ C$	R_{IO}	$> 10^{11}$	Ω
Isolation capacitance	$f = 1$ MHz	C_{IO}	0.8	pF
Safe input, output or total power	$\theta_{JA} = 86^\circ C/W$, $V_i = 5.5$ V, $T_J = 150^\circ C$, $T_A = 25^\circ C$	P_S	1430	mW
Safe input, output or supply current	$\theta_{JA} = 86^\circ C/W$, $V_i = 5.5$ V, $T_J = 150^\circ C$,	I_S	260	mA

	$T_A = 25\text{ }^{\circ}\text{C}$			
Maximum safe temperature		T_s	150	$^{\circ}\text{C}$
UL1577				
Per UL insulation voltage	$V_{\text{TEST}} = V_{\text{ISO}}$, $t = 60\text{ s}$ (qualification), $V_{\text{TEST}} = 1.2 \times V_{\text{ISO}}$, $t = 1\text{ s}$ (100% production test)	V_{ISO}	5000	V_{RMS}

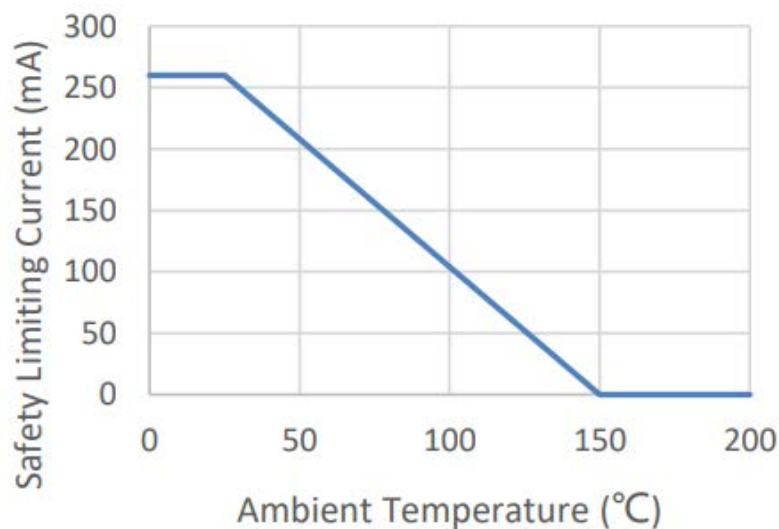


Figure 2-5. CMT1300 Thermal Derating Curve:

Relationship between Safety Limits Specified in DIN VDE V 0884-11 and Case Temperature

2.5 Electrical Characteristics

2.5.1 Electrical Characteristics: CMT1300D05

(VDD1 = 3.0V ~ 5.5V, VDD2 = 3.0V ~ 5.5V, INP = -50mV to +50mV, INN = GND1 = 0V, $T_A = -40^{\circ}\text{C}$ to 125°C . Unless otherwise stated, typical values are VDD1=5V, VDD2=3.3V, $T_A = 25^{\circ}\text{C}$)

Table 2-5-1. Recommended Operating Class^[1]

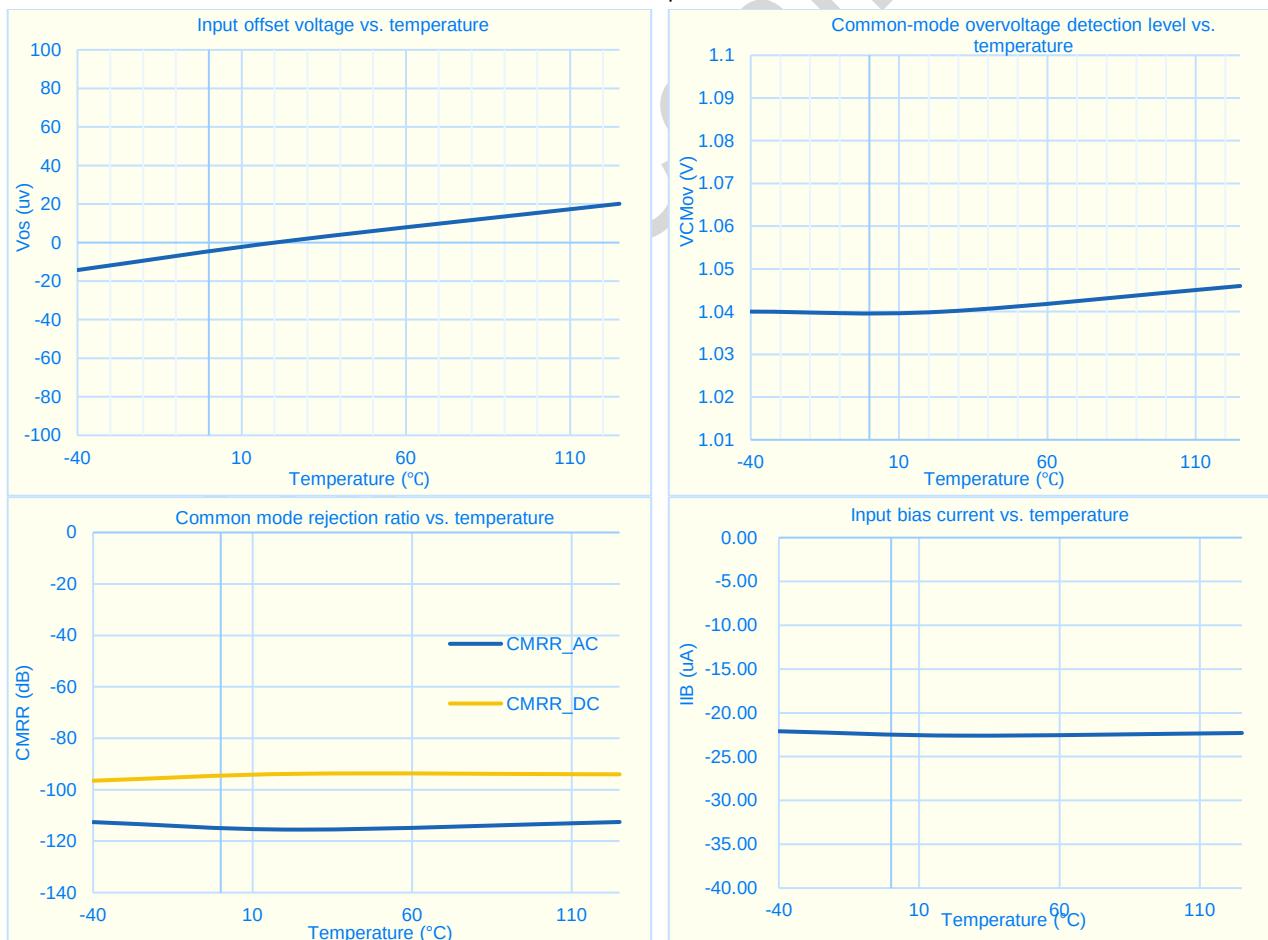
Parameter	Symbol	Min	Typ	Max	Unit	Description
Power Supply						
Side1 supply voltage	VDD1	3.0	5.0	5.5	V	
Side 2 supply voltage	VDD2	3.0	3.3	5.5	V	
Side1 power supply current	IDD1		11.4	15.1	mA	
Side 2 power supply current	IDD2		6.3	8.4	mA	
VDD1 undervoltage detection threshold voltage	VDD1 _{UV}	1.8	2.3	2.7	V	VDD1 decreased

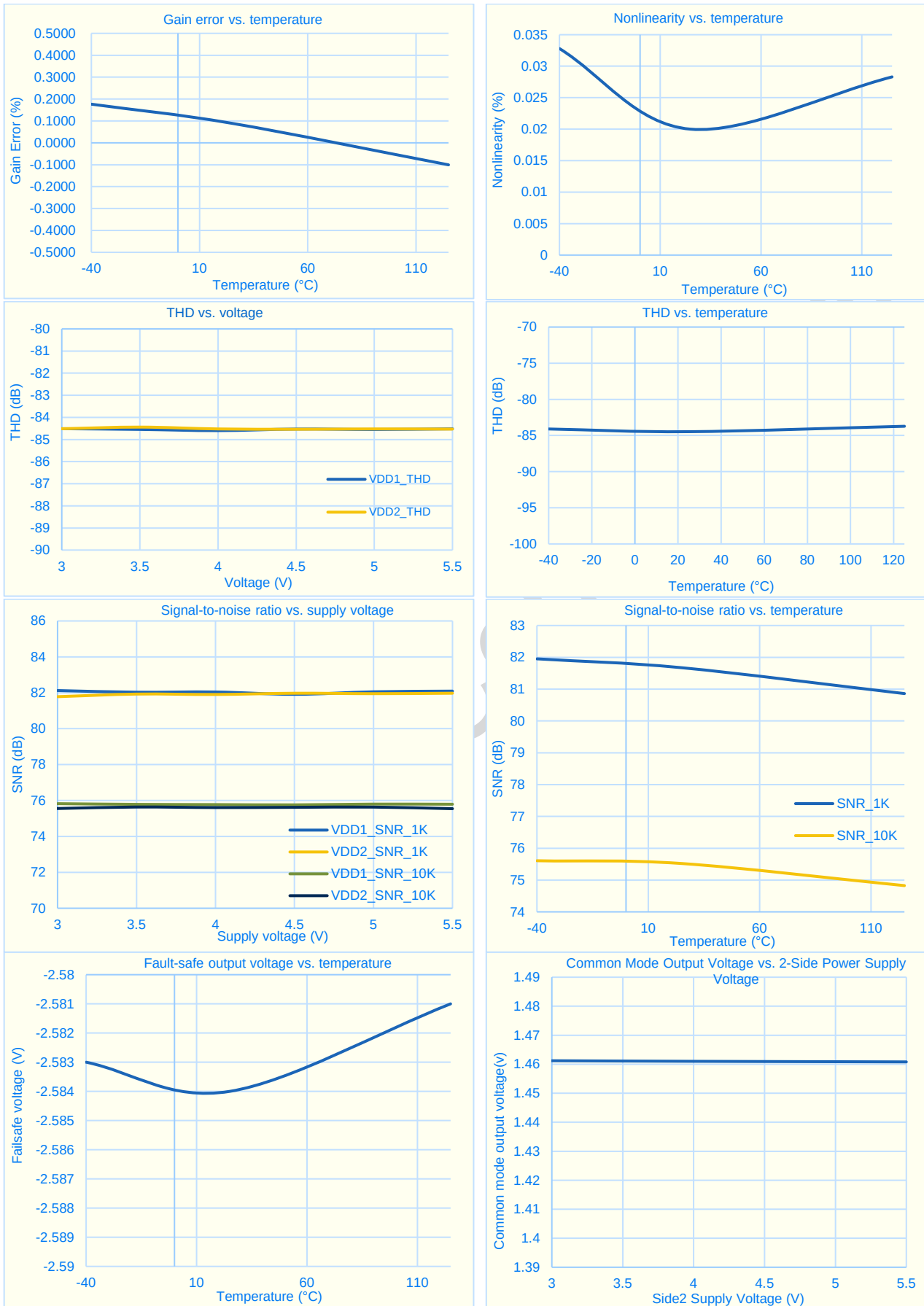
Analog Input						
Common mode overvoltage detection level	V _{CMov}	0.9			V	The detection level has a typical hysteresis of 96 mV
Input offset voltage	V _{OS}	-0.1	±0.01	0.1	mV	INP = INN = GND1
Input frequency offset drift	TCV _{OS}	-0.8	±0.15	1	μV/°C	
Common mode rejection ratio	CMRR _{dc}		-120		dB	INP = INN, f _{IN} = 0 Hz, V _{CM min} ≤ V _{IN} ≤ V _{CM max}
	CMRR _{ac}		-112		dB	INP = INN, f _{IN} = 10 kHz, V _{CM min} ≤ V _{IN} ≤ V _{CM max}
Single-ended input resistor	R _{IN}		4.75		KΩ	INN = GND1
Differential input resistor	R _{IND}		4.9		KΩ	
Input capacitance	C _I		2		pF	
Input bias current	I _{IB}	-29	-22	-14	μA	INP = INN = GND1, I _{IB} = (I _{IBP} + I _{IBN}) / 2
Input bias current drift	TCI _{IB}		±1.5		nA/°C	
Analog Output						
Calibrated gain			41		V/V	
Gain error	E _G	-0.3%	±0.05%	0.3%		
Gain error thermal drift	TCE _G	-50	±15	50	ppm/°C	
Nonlinearity		-0.04%	±0.01%	0.04%		
Nonlinear drift			±1		ppm/°C	
Total harmonic distortion	THD		-85		dB	f _{IN} = 10 kHz
Output noise			260		μV _{RMS}	INP = INN = GND1, BW = 100 kHz
signal-to-noise ratio	signal-to-noise ratio	80	84		dB	V _{IN} = 100 mVpp, f _{IN} = 1 kHz, BW = 10 kHz
			72		dB	V _{IN} = 100 mVpp, f _{IN} = 10 kHz, BW = 100 kHz
Common mode output voltage	V _{CMout}	1.39	1.44	1.49	V	
Fault protected differential output voltage	V _{FAILSAFE}		-2.6	-2.5	V	V _{CM} > V _{CMov} or VDD1 is missing
Output bandwidth	BW	250	310		kHz	
Power supply rejection ratio ⁽¹⁾	PSRR _{dc}		-118		dB	In DC state, PSRR and VDD1
	PSRR _{ac}		-116		dB	PSRR versus VDD1, 100 mV, and 10 kHz ripple
	PSRR _{dc}		-108		dB	In DC state, PSRR and VDD2
	PSRR _{ac}		-97		dB	PSRR versus VDD2, 100 mV, and 10 kHz ripple

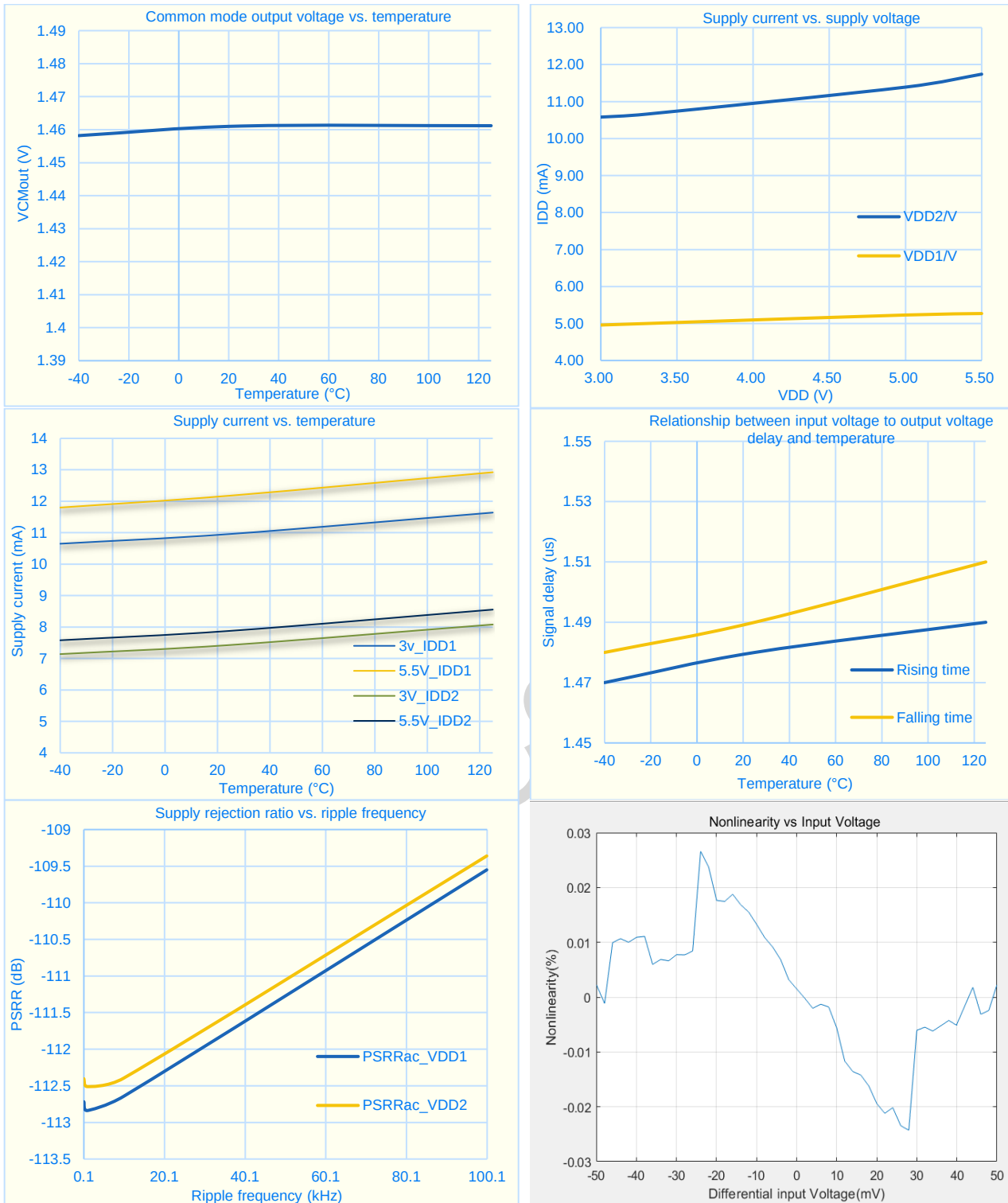
Output resistance	R_{OUT}		<0.2		Ω	
Output short-circuit current	$I_{OUT,OC}$		± 13		mA	
Common mode transient immunity	CMTI	100	150		kV/ μ s	Common mode transient immunity
Timing						
OUTP, OUTN rising time	t_r		1.3		μ s	
Falling time of OUTP, OUTN	t_f		1.3		μ s	
Signal delay from INP, INN to OUTP, OUTN (50%-50%)	t_{PD}		1.6	2.1	μ s	
Analog setting time	t_{AS}		0.5		ms	VDD1 steps up to 3.0 V, VDD2 ≥ 3.0 V to validate OUTP and OUTN, 0.1% setting.
(1) Reference input.						

2.6 Typical Characteristics

Unless otherwise stated, VDD1=5V, VDD2=3.3V, $T_A = 25^\circ\text{C}$, $C_L = 0$ pF.







2.7 Safety-related certification

Table 2-7. Safety-related Certifications

VDE	CSA	UL	CQC	TUV
DIN VDE V0884-11: 2017-01 (Patent Application)	IEC 60950-1, IEC 62368-1 and IEC 61010-1 (Patent Application)	Accredited under the UL 1577 Component Accreditation Program (Patent Application)	GB 4943.1-2022	EN 61010-1:2010 (3rd Edition) and EN 60950-1:2006/A2: 2013 (patent pending)
Certificate No.: pending	Master Contract No.: pending	Document No.: pending	Certificate Number: CQC23001382478	Customer ID Number: pending

3 Parameter Measurement Information

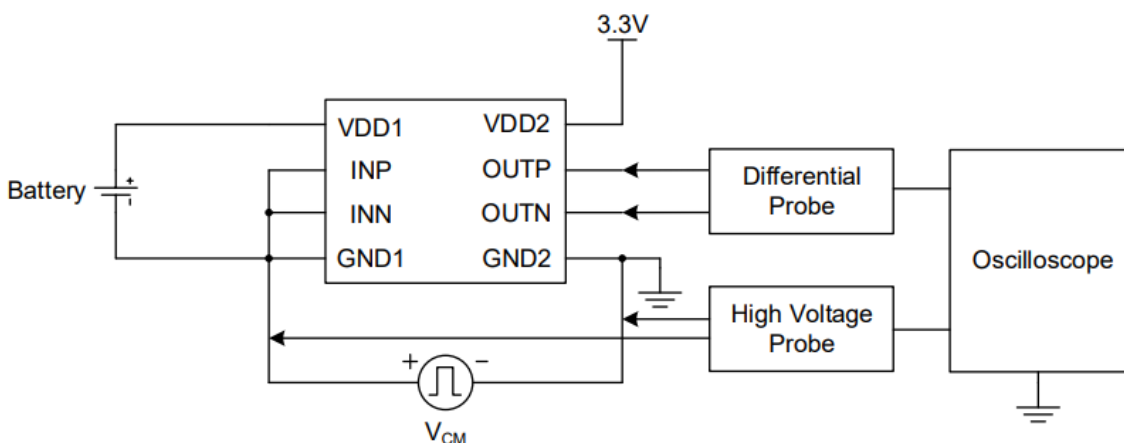


Figure 3-1. Common Mode Transient Immunity Test Circuit

4 Function Description

4.1 Overview

The CMT1300 is a high-performance enhanced isolation amplifier with fully differential input. The fully differential input structure is particularly suitable for high-voltage applications requiring electrical isolation, enabling efficient shunt current monitoring. The analog input is continuously sampled by a second-order $\Sigma\Delta$ modulator integrated in the device. Driven by an on-chip front-end fully differential amplifier, the modulator converts the analog input signal into a digital bitstream using an internal voltage reference and clock generator. The output of the modulator is transmitted by the driver (referred to as TX in the functional block diagram) across the isolation barrier that separates the voltages on Side 1 and Side 2 of the isolation. As shown in the functional block diagram, the received bit stream and clock are synchronized and processed by a fourth-order analog filter on Side 2, then output differentially.

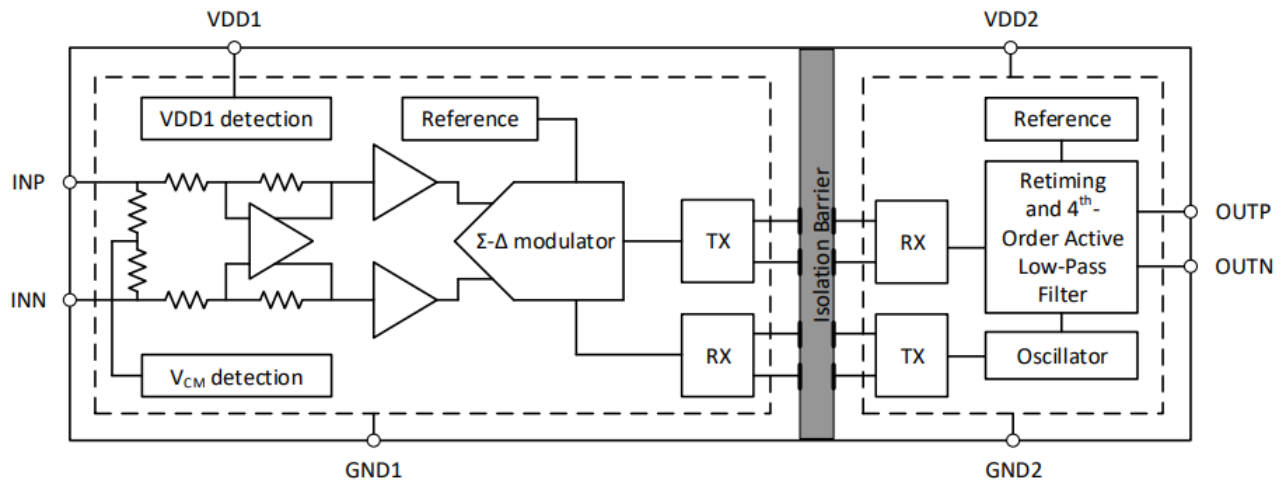


Figure 4-1. Functional Block Diagram

4.2 Analog Input

The analog input signal has 2 limits (V_{INP} and V_{INN}).

- If the input voltage exceeds the range from $GND1 - 6\text{ V}$ to $VDD1 + 0.5\text{ V}$, the internal electrostatic discharge (ESD) protection diodes of the device will turn on, and the input current must be limited to within 10 mA at this time.
- The linearity and noise performance of the device are guaranteed only when the analog input voltage remains within the specified full-scale range (FSR) and the specified common-mode input voltage range.

4.3 Analog Output

For the linear input range, the analog output of the CMT1300 has a fixed gain (41 for the CMT1300D05). If a full-scale input signal ($V_{IN} \geq V_{Clipping}$) is applied to the CMT1300, the analog output will be clamped (typically 2.45 V for positive clamping and -2.45 V for negative clamping).

In addition, the CMT1300 integrates several diagnostic measures and provides a fail-safe output to simplify system-level design. The fail-safe output is a negative differential output voltage that does not occur during normal device operation and is only activated under the following conditions:

- When VDD1 undervoltage is detected ($VDD1 < VDD1_{UV}$).

- When common mode input voltage overvoltage is detected ($V_{CM} > V_{CMOV}$).

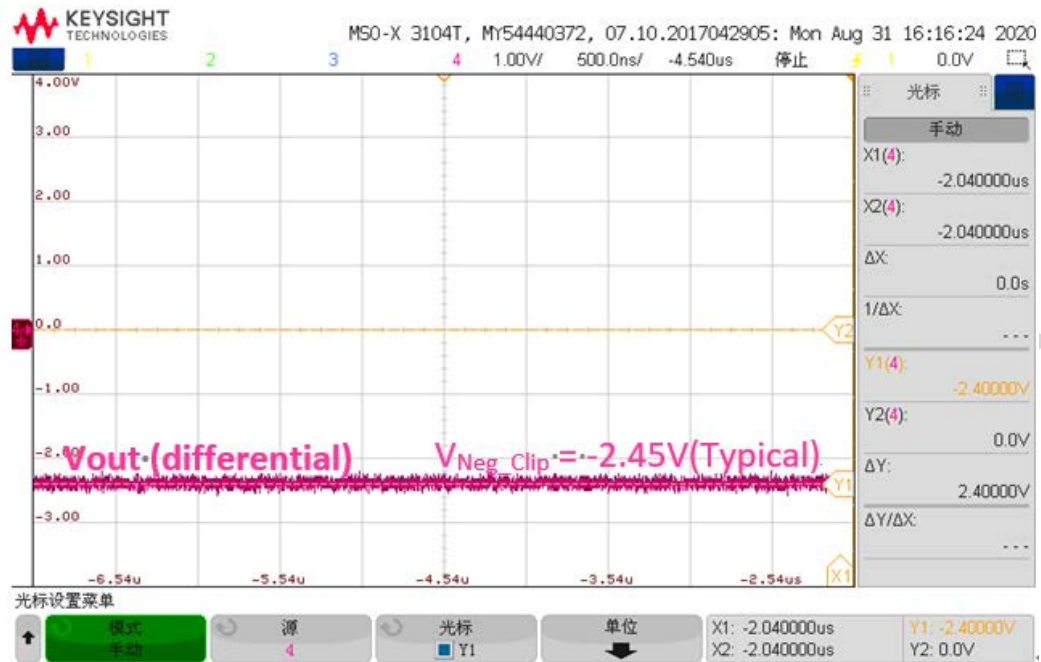


Figure 4-2. Typical Negative Clamp Output

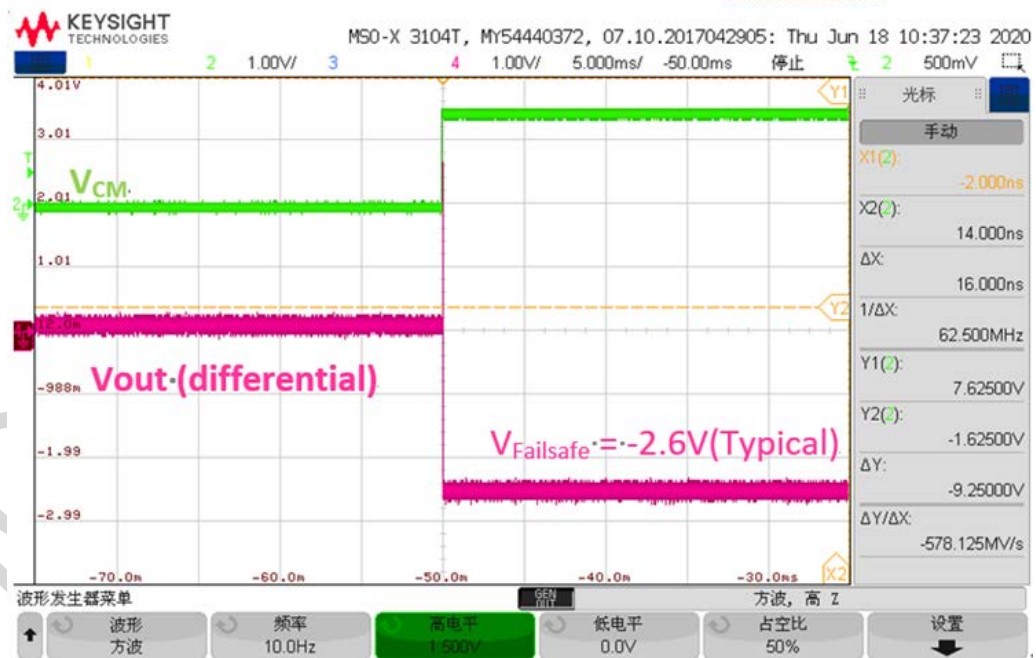


Figure 4-3. Typical Fail-Safe Output during Input Common-Mode Overvoltage

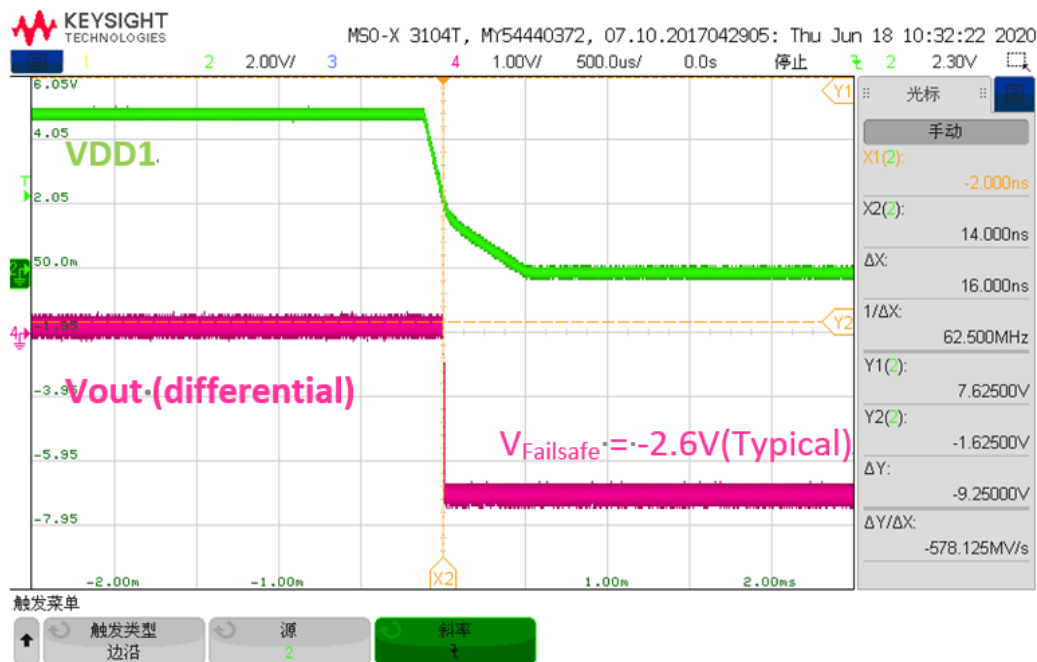


Figure 4-4. Typical fault-Safe Output when VDD1 is Under-voltage

5 Functional Description

5.1 Overview

CMT1300 is ideally suited for shunt resistor-based current sensing in high-voltage applications such as variable-frequency drives. A typical application circuit is shown in Figure 5-1.

The voltage across the shunt resistor R_{sense} is applied to the differential input terminals of the CMT1300 via an RC filter. The differential output of the isolation amplifier is converted to a single-ended analog output via an op-amp-based circuit. It is recommended to add resistors greater than 1 kΩ at the OUTP and OUTN pins to prevent output overcurrent. An analog-to-digital converter typically receives this analog output and converts it into a digital signal for processing by the controller.

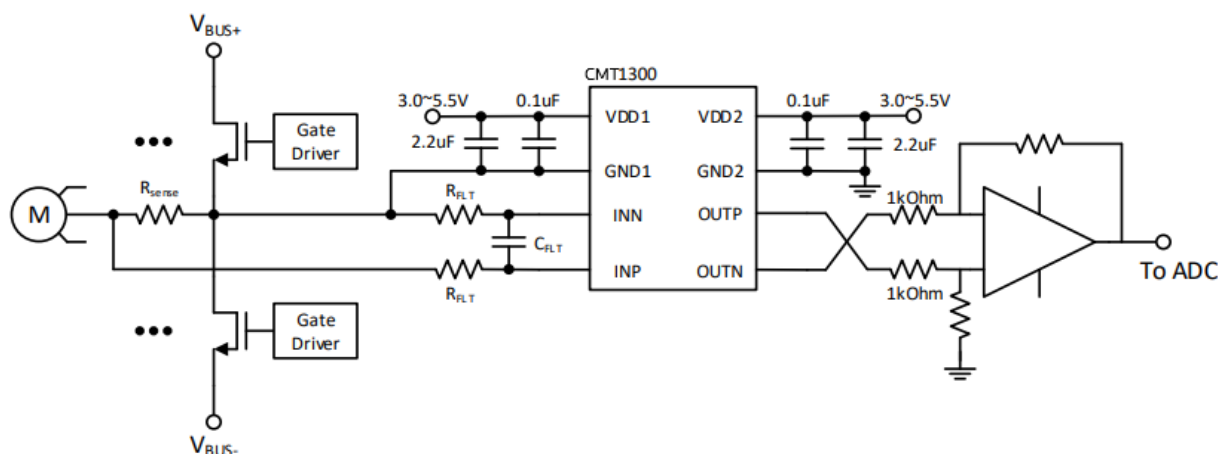


Figure 5-1. Typical Application Circuits in in-phase Current Sensing

5.2 Selection of Shunt Resistor

Selecting a specific shunt resistor typically requires a trade-off between minimizing power consumption and maximizing accuracy. A smaller detection resistor reduces power consumption, while a larger detection resistor can improve measurement accuracy by utilizing the full input range of the isolation amplifier.

Two additional factors should be considered when selecting the shunt resistor:

- The voltage drop caused by the rated current range must not exceed the recommended linear input voltage range:
 $V_{SHUNT} \leq FSR$.
- The voltage drop caused by the maximum allowable overcurrent must not exceed the input voltage that results in output clipping: $V_{SHUNT} \leq V_{Clipping}$.

5.3 PCB Layout

There are several key guidelines or considerations for optimizing performance in PCB layout:

- CMT1300 requires a 0.1 μF bypass capacitor connected between VDD1 and GND1, and another between VDD2 and GND2. This capacitor should be placed as close to the VDD pin as possible. For improved filtering, an additional capacitor with a value of 1–10 μF may be used.
- It is recommended to use Kelvin connection between the shunt resistor and the CMT1300. Since Kelvin connection is adopted, any voltage drop in the wires and leads will not affect the measured voltage.
- Place the shunt resistor close to the INP and INN input terminals, and maintain a symmetrical layout for the two connecting wires, connecting them closely and in parallel to the input terminals of the CMT1300. This minimizes the loop area of the connections and reduces the possibility of stray magnetic fields interfering with the measurement signal.

6 Order Information

Table 6-1. CMT1300 Ordering Information

Product Model	Isolation Level (V_{RMS})	Operating condition	Automotive grade	Package	Min Order Quantity
CMT1300D05	5000	-40 to 125°C	/	SOW8	1000

Please visit www.hoperf.com for more information about the product/product line.

Please contact sales@hoperf.com or your local sales representative for sales or pricing inquiries.

7 Package Information

The package information of CMT1300 is shown in the figure below.

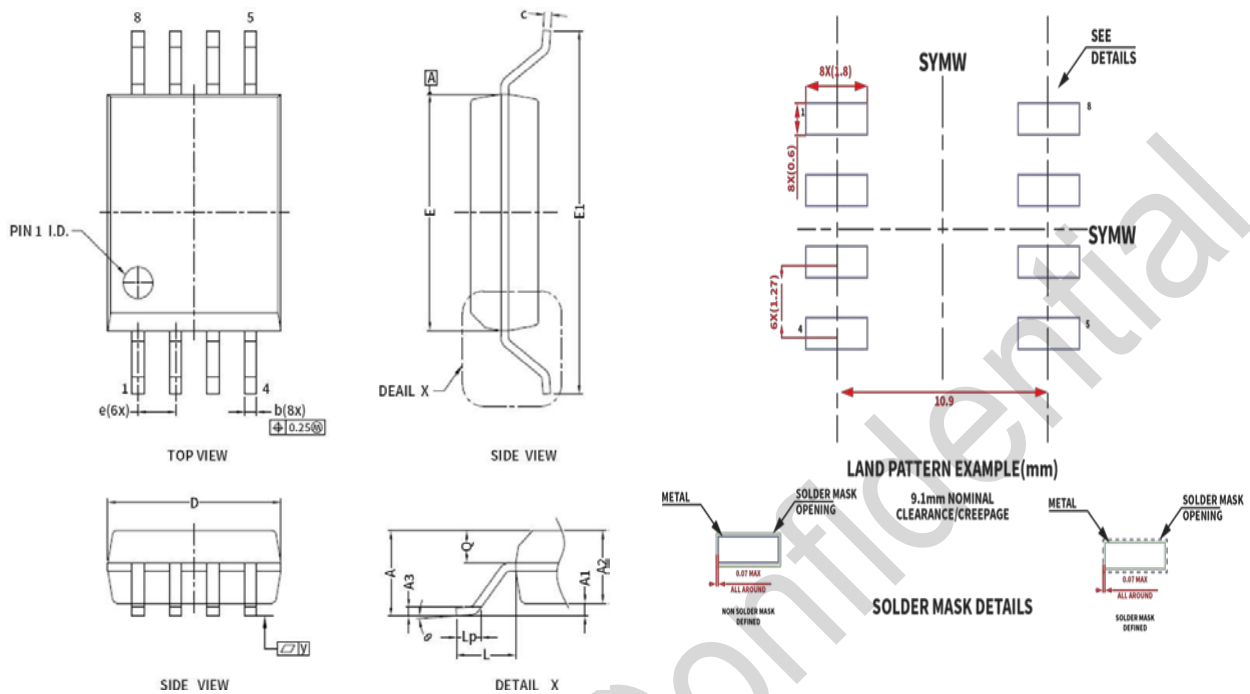
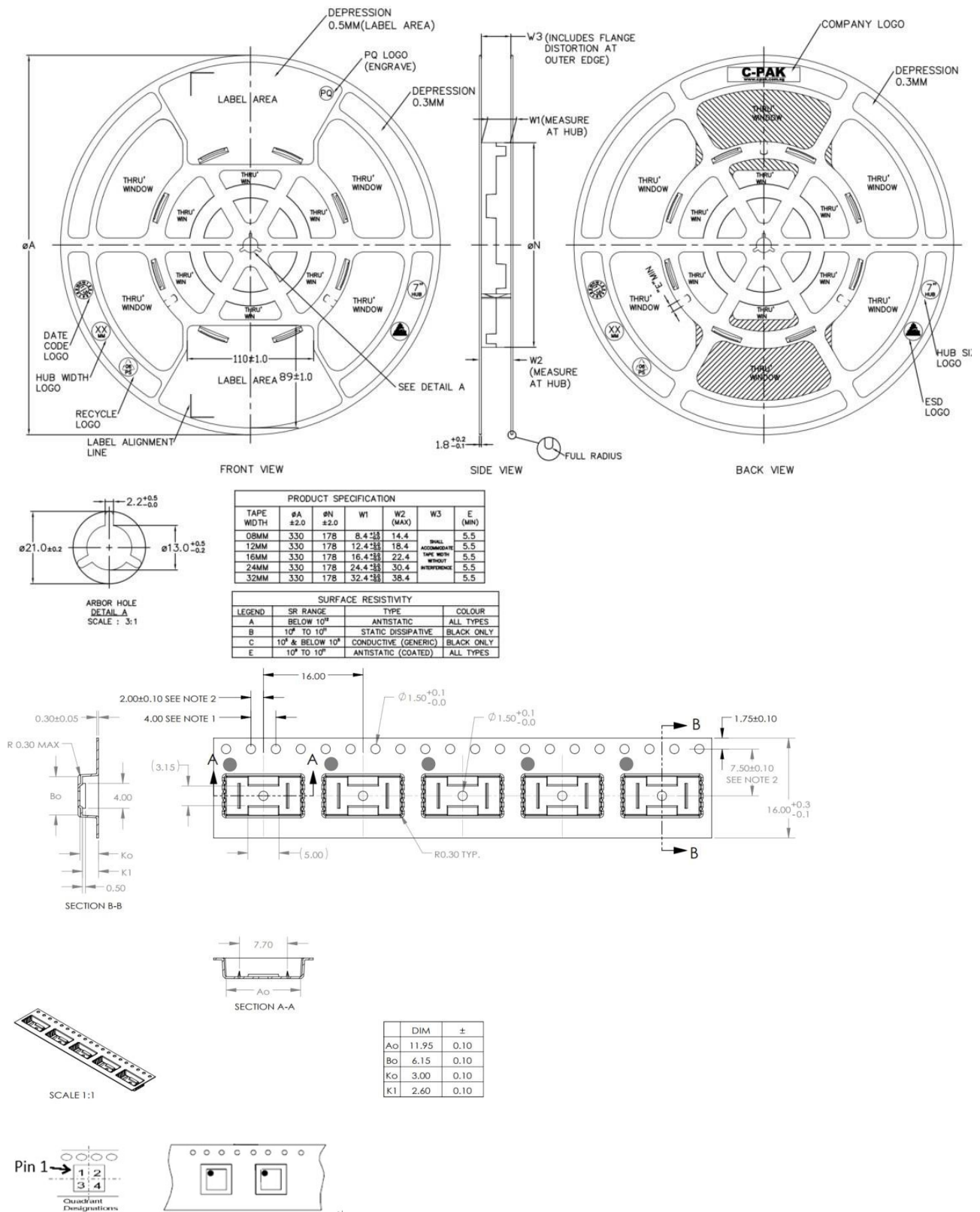


Figure 7-1. CMT1300 SOW8 Package

Table 7-1. CMT1300 SOW8 Package

Symbol	Size (mm)		
	Min	Typ	Max
A	-	-	2.80
A1	0.36	-	0.46
A2	2.20	2.30	2.40
A3	-	0.25	-
Q	0.97	1.02	1.07
b	0.31	0.41	0.51
c	0.13	-	0.33
D	5.75	5.85	5.95
E	7.40	7.50	7.60
E1	11.25	11.50	11.75
e	1.27 bsc		
L	2.00 bsc		
Lp	0.50	-	1.00
Y	-	0.10	-
θ	0	-	8°

8 Taping and Reel Information



9 Revision History

Table 9-1. Revision History

Version No.	Section	Description	Date
0.1	All	Initial version	2024-06-14
0.1	All	Review	2025-12-10

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10 Contact Information

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